

Astro Loyalty

REST API

Astro REST API functions for integration into the Astro Loyalty Frequent Buyer System, Offers, Loyalty Points and Marketing Suite

Authentication

Astro API Access Token

The Astro API uses access tokens for all backend API requests. To retrieve the access token you must pass the user credentials to the Astro API.

URL: <https://api.astroloyalty.com/api/json/token/>

Parameters:

- username – Astro Loyalty Username
- password – Astro Loyalty Password
- grant_type – Currently the only grant_type is password
- client_id – The client_id will be provided by Astro and should be passed with each token request

Returns Example

```
{
  access_token:D4mJFTGfScaDKco50p9CjyXjMJOfnNVqw0LGN9UqIDytakDwBG%2FfmYByzADPLiexLyW49QQFGsWomB9FNaxV1fcEJwhrqUYusdSvJlrS1J2vD0rYQURm
  token_type:bearer
  expires:1800
  created: 1519231484
}
```

Once you have retrieved the token, it is used in the Authorization header of all other requests.

Authorization:Bearer D4mJFTGfScaDKco50p9CjyXjMJOfnNVqw0LGN9UqIDytakDwBG%2FfmYByzADPLiexLyW49QQFGsWomB9FNaxV1fcEJwhrqUYusdSvJlrS1J2vD0rYQURm

Multi-Location API Support

The Astro API provides multi-location support via two different methods. The first method is just via basic authentication. If you use the Astro credentials from the different locations in a multi-location setup, Astro will record the API calls for the location whose credentials are passed to the token endpoint.

The second method will allow you to use a single set of credentials from the main location to obtain the token and then pass the internalLocationID variable with each subsequent API call. This method does require that Astro be configured with each sub location's internalLocationID in order to function properly, but it will allow for only a single set of credentials to be used for all locations in a multi-location setup.

Order Entry outside of the Astro Integration

It is important that any integration has the option of entering an order/sale that **DOES NOT** get sent to Astro. The reason for this requirement is due to the fact that there are multiple POS and eCommerce software companies integrated with Astro and the retailer has the option of using separate POS and eCommerce platforms. If the eCommerce platform has already sent the order, the POS needs to not duplicate it.

Customer Endpoints

customerStatus

The customerStatus API returns the customers current frequent buyer status with all open frequent buyer cards and purchases on those cards. Additionally, by setting the completed_cards flag, it will return all completed frequent buyer cards, the purchases on those cards and what was redeemed. It also returns loyalty points status, transactions and eligible rewards if applicable. The field email_status will be new (email address has not been verified), verified (email address has been verified) or opted_out (customer has unsubscribed from emails). The field sms_status will be new (customer has not opted in for SMS), opted_in (customer has opted in for SMS) or opted_out (customer has opted out for SMS). The customer_url can be used to direct the cashier to the Astro web app to view the customer within Astro (this should not be used for eCommerce Sites)

URL: <https://api.astroloyalty.com/api/json/customerStatus/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "completed_cards": [0 or 1] (optional – 0 for open cards, 1 for completed cards)
 "astro_mfg_program_id": [astro program id] (option – will narrow astroCardData to requested program)
}

Return Example

```
{
  status:100
  status_message:Success
  returnData: {
    customerID:11
    astro_customer_id: 123123
    first_name:Mark
    last_name:Keeran
    email_address: mkeeran@astroloyalty.com
    phone: 5555555555
    address: 1 Main St
    city: Fayetteville
    state: AR
    zip: 72703
    email_status: new
    sms_status: new
    customer_url: https://secure.astroloyalty.com/api/customer_redirect.php?rd=[redirect]
    linked_customerIDs: [
      {
        11
      }
    ]
    astroCardData: [
      {
        astro_program_id:72
      }
    ]
  }
}
```

```
astro_program_source:manufacturer
astro_program_title:Monster Naturals Canine Large
astro_mfg_name:Monster Naturals
astro_card_id:26850
astro_card_status:open
cardPurchases: [
    {
        imported_transaction_id: 1234
        astro_transaction_id: 31211
        item_code:041693221211
        item_description:Monster Adult Anhovy & Sardine
        purchase_date: 2018-02-20
    }
    {
        imported_transaction_id:1255
        astro_transaction_id: 31212
        item_code:041693221212
        item_description:Monster Adult Salmon
        purchase_date:2018-01-26
    }
]
cardFreeGoods:[
    {
        astro_reward_id: 12311
        astro_item_id: -1
        item_description:Any Eligible Program Item
        free_qty: 1
        redeemed_on:null
    }
]
astroOfferRewards [
    {
        offer_title: Monster Naturals - $2 Instant Rebate
        offer_type: rebate
        rebate_amount: 2
        offer_reward_id: 12312
        offer_expires: 2019-08-31
    }
    {
        offer_title: Monster Naturals – Free Treats
        offer_type: free_good
        offer_reward_id: 123111
        offer_expires: 2019-8-31
    }
]
astroPointsBalance: 525
astroPointsTransactions:[
    {
```

```
    astro_points_transaction_id: 131881
    pointsDescription: Earn 1 point per dollar
    pointsDate: 2018-11-05
    pointsQty: 25
    totalPoints: 25
  }
  {
    astro_points_transaction_id: 135871
    pointsDescription: Earn 1 point per dollar
    pointsDate: 2018-11-24
    pointsQty: 500
    totalPoints: 500
  }
]
astroPointsEligibleRewards:[
  {
    astro_points_reward_id: 1214
    pointsRewardTitle: Earn 300 points, Get $15 off
    pointsRequired: 300
    reward_type: rebate
    reward_value: 15
  }
]
}
}
```

customerRewardStatus

The customerRewardStatus API returns the customers current reward status for Frequent Buyer, Offers and Loyalty Points. If the customer is eligible for free goods, the eligible items will be listed.

URL: <https://api.astroloyalty.com/api/json/customerRewardStatus/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
}

Return Example

```
{
  status:100
  status_message:Success
  returnData: {
    customerID:11
    astro_customer_id: 123123
    first_name:Mark
    last_name:Keeran
    email_address: mkeeran@astroloyalty.com
    phone: 5555555555
    address: 1 Main St
    city: Fayetteville
    state: AR
    zip: 72703
    customer_url: https://secure.astroloyalty.com/api/customer_redirect.php?rd=[redirect]
    linked_customerIDs: [
      {
        11
      }
    ]
    frequentBuyerRewards: [
      {
        astro_card_id:26850
        astro_program_id:72
        astro_program_title:Monster Naturals Canine Large
        astro_reward_id:3094383
        earned_on: 2022-07-01
        cardFreeGoods:[
          {
            item_id: 3911
            item_description: Monster Naturals Pork & Beef
            item_code: 12331234123
          }
        ]
      }
    ]
    astroOfferRewards [
      {
```

```
offer_title: Monster Naturals - $2 Instant Rebate
offer_type: rebate
rebate_amount: 2
offer_reward_id: 12312
offer_expires: 2019-08-31
}
{
offer_title: Monster Naturals – Free Treats
offer_type: free_good
offer_reward_id: 123111
offer_expires: 2019-8-31
offerFreeGoods [
    {
        item_id: 3121
        item_description: Monster Chicken Treat
        item_code: 1234441233
    }
]
}
]
astroPointsBalance: 525
astroPointsEligibleRewards:[
    {
        astro_points_reward_id: 1214
        pointsRewardTitle: Earn 300 points, Get $15 off
        pointsRequired: 300
        reward_type: rebate
        reward_value: 15
    }
]
}
}
```

searchCustomer

The searchCustomer API allows you to search for a customer to the Astro platform by Email Address or Phone.

URL: <https://api.astroloyalty.com/api/json/searchCustomer/>

Parameters:

- jsonData = {
 "email_address": [customer email address]
 "phone": [customer phone]
}

Return Example

```
{  
  status: 100  
  status_message: Success  
  returnData: {  
    astro_customer_id: 12345  
    first_name: Mark  
    last_name: Smith  
  }  
}
```

searchCustomerByName

The searchCustomerByName API allows you to search for a customer to the Astro platform by first or last name. It will return an array of matches. **It is very important NOT to use this for an eCommerce site.**

URL: <https://api.astroloyalty.com/api/json/searchCustomerByName/>

Parameters:

- jsonData = {
 "first_name": [customer first name]
 "last_name": [customer last name]
}

Return Example

```
{  
  status: 100  
  status_message: Success  
  returnData: {  
    customerArray: [  
      {  
        astro_customer_id: 12345  
        first_name: Mark  
        last_name: Smith  
      }  
    ]  
  }  
}
```


linkCustomer

The linkCustomer API allows you to link an existing customer in Astro platform to a customer in the POS software.

URL: <https://api.astroloyalty.com/api/json/linkCustomer/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "astro_customer_id":[astro customer id]
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: {  
    astro_customer_id: 12345  
  }  
}
```

addCustomer

The addCustomer API allows you to add a customer to the Astro platform and link it to your internal customer id within your POS software.

URL: <https://api.astroloyalty.com/api/json/addCustomer/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "first_name":[first_name]
 "last_name":[last_name]
 "email_address":[email_address] (optional)
 "address":[address1] (optional)
 "city":[city] (optional)
 "state":[state] (optional)
 "zip":[zip] (optional)
 "phone":[phone] (optional)
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: {  
    astro_customer_id: 12345  
  }  
}
```

updateCustomer

The updateCustomer API allows you to update an existing customer's information in the Astro platform.

URL: <https://api.astroloyalty.com/api/json/updateCustomer/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "first_name":[first_name]
 "last_name":[last_name]
 "email_address":[email_address] (optional)
 "address":[address1] (optional)
 "city":[city] (optional)
 "state":[state] (optional)
 "zip":[zip] (optional)
 "phone":[phone] (optional)
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: {  
    astro_customer_id: 12345  
  }  
}
```

unlinkCustomer

The unlinkCustomer API allows you to remove the association of an existing customer in Astro platform to a customer in the POS software.

URL: <https://api.astroloyalty.com/api/json/unlinkCustomer/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "astro_customer_id":[astro customer id]
}

Return Example

```
{  
  status:100  
  status_message:Success  
}
```

Product End Points

eligiblePurchaseItems

The eligiblePurchaseItems API allows you to obtain a list of eligible UPCs for frequent buyer programs and offersthe retailer is enrolled in. It could return a lot of data, so it should be used judiciously as not to slow down the point of sale transactions. It is possible to pass a single UPC to check its eligibility, but if no UPC is passed, all products are returned. It is suggested that this be used in a sync scenario that the retailer can run once in a while to mark items in their inventory that are Astro eligible. It is not required, however, and any UPC can be passed via addTransaction and the API will return a status of Not Found if the item is not in the Astro platform. If item_code is passed, this will return programs and offers associated.

URL: <https://api.astroloyalty.com/api/json/eligiblePurchaseItems/>

Parameters:

- jsonData = {
 "item_code": [optional – the UPC to check against the Astro product database]
 "item_code_batch": [optional - an array of UPCs to check against the Astro product database]
}

Return Example

```
{
  status:100
  status_message:Success
  returnData: [
    {
      astro_mfg_id: 101
      astro_mfg_name: Astro MFG
      astro_item_id: 3211
      item_code: 01230001000
      item_description: Monster Naturals Turkey and Salmon 10lb
      available_programs {
        astro_mfg_program_id: 100
        astro_mfg_program_title: Monster Naturals Frequent Buyer Program
        astro_mfg_program_card: Caninie Medium
        astro_mfg_program_type: frequent_buyer
        astro_new_card_cutoff_date: NULL (if date is returned no new cards can be
started after date)
        in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
      } * (available_programs only returned if item_code is passed)
    }
    {
      astro_mfg_id: 101
      astro_mfg_name: Astro MFG
      astro_item_id: 3212
      item_code: 01230001001
      item_description: Monster Naturals Pork 10lb
      available_programs {
```

```

        astro_mfg_program_id: 100
        astro_mfg_program_title: Monster Naturals Frequent Buyer Program

        astro_mfg_program_card: Caninie Medium
        astro_mfg_program_type: frequent_buyer
        in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
        astro_new_card_cutoff_date: NULL (if date is returned no new cards can be
started after date)
    }
}
]
}

```

eligibleRewardItems

The eligibleRewardItems API returns a list of eligible free goods for an astro_reward_id provided by the customerStatus API for a full astro frequent buyer card.

URL: <https://api.astroloyalty.com/api/json/eligibleRewardItems/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "astro_reward_id": [astro reward id]
 }

Return Example

```

{
  status: 100
  status_message: Success
  returnData: {
    customerID: 112233
    astro_reward_id: 12345
    astro_reward_source: manufacturer
    customer_address_required: true
    customer_email_required: true
    eligibleItemArray[
      {
        astro_item_id: 44
        item_upc: 041693239101
        item_description: MN Chicken Meal 15lbs
      }
      {
        astro_item_id: 45
        item_upc: 041693239102
        item_description: MN Lamb Meal 15lbs
      }
    ]
  }
}

```

} }

Frequent Buyer End Points

addTransaction

The addTransaction API allows you to add a customer purchase to the Astro platform. Astro will determine the correct program and return a confirmation of the Astro Program ID and Title. You can add a single transaction with a multiple quantity but all Astro transactions will be linked to that internal transaction id and if that internal transaction id is deleted, all Astro transactions linked to it will be deleted as well, provided they are not on a redeemed frequent buyer card. Likewise, if necessary, you can add an option transaction date if the need arises to post a transaction that was not earlier recorded.

URL: <https://api.astroloyalty.com/api/json/addTransaction/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "transactionID": [internal unique transaction id]
 "saleID": [optional – internal saleID]
 "item_code": [UPC of the purchased item]
 "item_qty": [optional, if not present 1 is assumed]
 "item_transaction_date": [optional, if not present today's date is assumed]
}

Return Example

```
{
  status: 100
  status_message: Success
  returnData: [
    {
      astro_transaction_id: 12345
      astro_program_id: 131
      astro_program_title: MFG Canine Large Card
    }
    {
      astro_transaction_id: 12346
      astro_program_id: 131
      astro_program_title: MFG Canine Large Card
    }
  ]
}
```

addTransactionBatch

The addTransactionBatch API allows you to add a multiple purchase items for a single customer to the Astro platform. Astro will determine the correct program and return a confirmation of the Astro Program ID and Title.

URL: <https://api.astroloyalty.com/api/json/addTransactionBatch/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "saleID": [optional – internal sale id]
 "transactions": [
 {
 "transactionID": [internal unique transaction id]
 "item_code": [UPC of the purchased item]
 "item_qty": [optional, if not present 1 is assumed]
 "item_transaction_date": [optional, if not present today's date is assumed]
 }
 {
 "transactionID": [internal unique transaction id]
 "item_code": [UPC of the purchased item]
 "item_qty": [optional, if not present 1 is assumed]
 "item_transaction_date": [optional, if not present today's date is assumed]
 }
]
}

Return Example

```
{
  status: 100
  status_message: Success
  returnData: [
    {
      transaction_status: 100
      transaction_status_message: Success
      astro_transaction_id: 12345
      astro_program_id: 131
      astro_program_title: MFG Canine Large Card
    }
    {
      transaction_status: 100
      transaction_status_message: Success
      astro_transaction_id: 12346
      astro_program_id: 131
      astro_program_title: MFG Canine Large Card
    }
  ]
}
```

removeTransaction

The removeTransaction API allows you to remove a previously added customer purchase to the Astro platform. Astro will ensure the transaction is not on a completed and redeemed card before removing. If the transaction had a multiple quantity, all Astro transactions linked to that internal transaction id will be removed unless an item_qty or astro_transaction_id are specified. If item_qty is specified, it will remove that qty starting from the last astro_transaction_id working backwards. If astro_transaction_id is specified it will only remove that transaction.

URL: <https://api.astroloyalty.com/api/json/removeTransaction/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "transactionID": [internal unique transaction id]
 "item_qty": [quantity to remove – optional]
 "astro_transaction_id": [the astro_transaction_id provided in getCustomerStatus – optional]
}

Return Example

```
{
  status:100
  status_message:Success
  returnData:[
    {
      transactionID: 312239991
      astro_transaction_id: 12345
      transactionDeleted: true
    }
    {
      transactionID: 312239991
      astro_transaction_id: 12346
      transactionDeleted: true
    }
  ]
}
```


removeSale

The removeSale API endpoint allows you to remove all transactions on a sale or on a particular date as long as none of those transactions are tied to a redeemed reward. If any of them are, you will need to remove the redemption prior to removing the sale.

URL: <https://api.astroloyalty.com/api/json/removeSale/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "saleID": [internal saleID – optional but this or saleDate must be included]
 "saleDate": [saleDate – optional but this or saleID must be included]
}

Return Example

```
{
  status:100
  status_message:Success
  returnData:[
    {
      transactionID: 312239991
      astro_transaction_id: 12345
      transactionDeleted: true
    }
    {
      transactionID: 312239991
      astro_transaction_id: 12346
      transactionDeleted: true
    }
  ]
}
```

checkRedemptionEligibility

The checkRedemptionEligibility API will tell you if a UPC for a customer is eligible as a free item.

URL: <https://api.astroloyalty.com/api/json/checkRedemptionEligibility/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "item_code": [the items UPC]
}

Return Example

```
{  
  status: 100  
  status_message: Success  
  returnData: {  
    isEligible: true (will return false if not)  
    astro_reward_id: 12345  
    astro_item_id: 2313  
    program_type: frequent_buyer (will be frequent_buyer or offer)  
    redemption_endpoint: addRedemption (addRedemption or addOfferRedemption)  
  }  
}
```

addRedemption

The addRedemption API allows you to add a customer redemption to the Astro platform. To obtain the eligible astro_item_ids, first call eligibleRewardItems. Some MFGs require customer address or email to complete the redemption. If that is the case, eligibleRewardItems will provide that information and the customer information will be required in the API call. If not, the customer information can be left blank.

URL: <https://api.astroloyalty.com/api/json/addRedemption/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "astro_reward_id": [astro reward id]
 "astro_item_id": [astro item id]
 "customer_email_address": [customer's email address]
 "customer_address": [customer's mailing address]
 "customer_city": [customer's mailing city]
 "customer_state": [customer's mailing state]
 "customer_zip": [customer's postal code]
}

Return Example

```
{  
  status: 100  
  status_message: Success  
}
```

removeRedemption

The removeTransaction API allows you to remove a previously added customer redemption to the Astro platform. Astro will ensure the redemption has not fulfilled before removing.

URL: <https://api.astroloyalty.com/api/json/removeRedemption/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "astro_reward_id":[astro_reward_id]
}

Return Example

```
{  
    status:100  
    status_message:Success  
}
```

Astro Offers Endpoints for POS without capability to run Offers locally

The Astro API contains the necessary endpoints to allow the POS/eCommerce software to use Astro's Offer engine to run offers in real time during the checkout process. One main difference between offers and frequent buyer is that it is important that the offer transactions are added during the sales process so the free goods can be applied to the existing sale. Offer participation must occur in a single visit. As transactions are added, if a reward is earned it will be included in the return data. If your POS/eCommerce software has the ability to run offers such as Buy 1 Get 1, Buy 2 Get 1 and rebates, you may be able to utilize the second set of Offer endpoints that will allow you to run those locally and only communicate the redemptions to Astro. **Some offers, per terms and conditions, are not allowed to run on eCommerce platforms. If `in_store_only` is set to 1, you must not run the offer on an eCommerce platform.**

listOffers

The listOffers API endpoint provides a list of active offers that the retailer has enrolled in. It provides the title, description, start and end dates and program image.

URL: <https://api.astroloyalty.com/api/json/listOffers/>

Parameters: None

Return Example

```
{
  astro_mfg_id: 15
  astro_mfg_name: Monster Naturals
  astro_program_id: 114
  astro_program_title: Monster Naturals | Free Treats with Purchase of Bag
  astro_program_long_description: Get a free bag of dog treats with a purchase of any 15# bag of
dog food
  astro_program_start_date: 2019-08-01
  astro_program_end_date: 2019-08-31
  astro_program_image: https://secure.astroloyalty.com/display\_image.php?ec=sdfk2377623jhh
  in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
}
```

addOfferTransaction

The addOfferTransaction API allows you to add a customer purchase to the Astro platform for an eligible offer. Astro will determine the correct program and return a confirmation of the Astro Program ID and Title. You can add a single transaction with a multiple quantity but all Astro transactions will be linked to that internal transaction id and if that internal transaction id is deleted, all Astro transactions linked to it will be deleted as well, provided they are not on a redeemed offer.

It is possible that a product could be on two offers simultaneously. Though rare, if that is the case, Astro will let you know (status 770, Offer Program ID Required) and you will need to resubmit and include the offer you wish the transaction to go towards. Additionally, unlike Frequent Buyer, Offers often have eligibility limits, either in total or per customer. If the eligibility limit has been reached, you will receive a status of 755, Max Eligibility Reached. Finally, almost all offers require the purchases be made in a single visit. So if it is Buy2, Get1 and the customer only purchases 1 this visit, the next visit that purchase will not count towards the offer. **Rebates are cumulative, note that in the return data below.**

URL: <https://api.astroloyalty.com/api/json/addOfferTransaction/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "transactionID": [internal unique transaction id]
 "saleID": [option – internal saleID]
 "item_code": [UPC of the purchased item]
 "item_qty": [optional, if not present 1 is assumed]
 "astro_offer_id": [only required if item is on more than one active offer]
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: [  
    {  
      astro_transaction_id: 12345  
      astro_program_id: 131  
      astro_program_title: MFG $2 Instant Rebate  
      program_type: rebate  
      rebate_amount: 2.00 (only on rebates)  
      astro_reward_id: 2423111 (only on rebates)  
    }  
    {  
      astro_transaction_id: 12346  
      astro_program_id: 131  
      astro_program_title: MFG $2 Instant Rebate  
      program_type: rebate  
      rebate_amount: 4.00 (Notice rebate amount is doubled)  
      astro_reward_id: 2423111 (but reward id is the same, so still $2 per item)  
    }  
  ]  
}
```

removeOfferTransaction

The removeOfferTransaction API allows you to remove a previously added customer offer purchase to the Astro platform. Astro will ensure the transaction is not on a completed and redeemed offer before removing. If the transaction had a multiple quantity, all Astro transactions linked to that internal transaction id will be removed.

URL: <https://api.astroloyalty.com/api/json/removeOfferTransaction/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "transactionID":[internal unique transaction id]
}

Return Example

```
{
  status:100
  status_message:Success
  returnData:[
    {
      transactionID: 312239991
      astro_transaction_id: 12345
      transactionDeleted: true
    }
    {
      transactionID: 312239991
      astro_transaction_id: 12346
      transactionDeleted: true
    }
  ]
}
```

addOfferRedemption

The addOfferRedemption API allows you to add a customer redemption to the Astro platform. In the case of a rebate, no astro_item_id is required. If the offer is a free good offer, you will need to provide the astro_item_id to redeem the offer. To obtain the eligible astro_item_ids, first call eligibleRewardItems. This is the same endpoint that is used to obtain eligible items for Frequent Buyer. Some MFGs require customer address or email to complete the redemption. If that is the case, eligibleRewardItems will provide that information and the customer information will be required in the API call. If not, the customer information can be left blank.

URL: <https://api.astroloyalty.com/api/json/addOfferRedemption/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "astro_reward_id": [astro reward id]
 "astro_item_id": [astro item id – not required for rebates]
 "customer_email_address": [customer's email address]
 "customer_address": [customer's mailing address]
 "customer_city": [customer's mailing city]
 "customer_state": [customer's mailing state]
 "customer_zip": [customer's postal code]
}

Return Example

```
{  
    status:100  
    status_message:Success  
}
```

*To remove an offer redemption, simply use the same removeRedemption endpoint that is used for Frequent Buyer redemptions.

Astro Offers Endpoints for POS with capability to run Offers locally

The Astro API allows for the POS/eCommerce software to run Astro offers locally and only transmit the redemptions to Astro at the end of the sale. If the software has this capability, it is the more efficient method to handle Astro offers. Astro provides an end point that will return the complete setup of a single offer to allow that offer to be automatically imported into the POS/eCommerce software, thereby helping to ensure accurate setup and reduce the requirements on the retailer to participate in the offer. The retailer simply needs to enroll in the offers in Astro and they will be available via the API. **Some offers, per terms and conditions, are not allowed to run on eCommerce platforms. If in_store_only is set to 1, you must not run the offer on an eCommerce platform.**

There are few different calculation types for Astro Offers. They are as follows:

- Rebate – These are instant rebates in which the customer receives dollars off an item purchase.
- BuyXGetY – This is a free good offer in which the customer receives a free item for purchasing an item. In this type, it is a one to one relationship so the customer buys Item A and gets Item C free or buys Item B and gets Item D free.
- BuyXGetYAnyProduct – This is a free good offer in which the customer buys any of Item A, B or C and can then choose any of Item A, B or C free. In some cases, these do have a lesser or equal value redemption restriction so the customer may not redeem an item that is more expensive than the item they purchased. In the event of that restriction, item values are provided by Astro.
- BuyXGetYSameProduct – This is a free good offer in which the customer buys any of Item A, B or C and gets the same item free. If the customer purchases Item B, they receive an additional Item B for free.
- BuyAnyXGetYAnyOtherY – This is a free good offer in which the customer buys any of Item A, B or C and receives any of Item D, E or F for free.
- BuyXandYGetRebate – This is a rebate offer that requires the pet owner to purchase a combination of items to receive a rebate on one of the items. The program type has purchase slots, so the pet owner is required to purchase 1 (or more) items from each purchase slot. If they do that, they will receive a rebate on one of the items.
- FreePunch – This is a unique type of offer that has no reward. Instead when an item on the free punch offer is purchased, a second punch will be added to the frequent buyer card. This one will be a bit trickier to run locally. Essentially if they are enrolled in this program and purchase an item, you will need to either utilize addTransaction TWICE in order to give them their free punch or, in realtime, call the addOfferTransaction (again, there is no redemption for this program type so you cannot utilize addOfferRedemptionOnly)

Astro Offers also have some other restrictions that are not typical of Frequent Buyer programs. All Offers have a start and end date and are only valid between those dates. Some offers run for only a day or a week but most are typically monthly offers. If the POS attempts to send a redemption for an Offer that is outside of those dates, it will be rejected by Astro. Astro Offers also have individual customer redemption restrictions. For example, limit 2 per customer. These limits will be provided and must be followed. If redemptions are attempted past the max individual limit, they will be rejected. Since the POS is running these offers locally and only submitting after the customer completes the sale, be sure to keep track of these customer limits and enforce them through the length of the offer. If the max_individual_eligibility parameter is set to -1, there is no limit on redemptions.

listOffers

The listOffers API endpoint provides a list of active offers that the retailer has enrolled in. It provides the title, description, start and end dates and program image.

URL: <https://api.astroloyalty.com/api/json/listOffers/>

Parameters: None

Return Example

```
{
  astro_mfg_id: 15
  astro_mfg_name: Monster Naturals
  astro_program_id: 114
  astro_program_title: Monster Naturals | Free Treats with Purchase of Bag
  astro_program_long_description: Get a free bag of dog treats with a purchase of any 15# bag of
dog food
  astro_program_start_date: 2019-08-01
  astro_program_end_date: 2019-08-31
  astro_program_image: https://secure.astroloyalty.com/display\_image.php?ec=sdfk2377623jhh
  in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
}
```

mfgProgramDetails

The mfgProgramDetails API returns the information needed to reconstruct an offer locally within the POS/eCommerce software. All the parameters needed, including skus, values, start and end dates, etc are provided.

URL: <https://api.astroloyalty.com/api/json/mfgProgramDetails/>

Parameters:

- jsonData = {
 "astro_program_id": [astro_program_id from listOffers]
}

Return Examples:

Rebate – with a rebate, there are only purchase items and each has it's own rebate value. That is the dollars back the POS will give the customer for the purchase of the item. The upc purchased is also the UPC used for the redemption API call.

```
{
  astro_mfg_id: 51
  astro_mfg_name: Primal Pet Foods
  astro_mfg_program_title: Primal | Dollars off New Market Mix
  astro_program_type: offer
  program_start_date: 2020-07-01
  program_end_date: 2020-07-31
  program_calculation_type: Rebate
  redemption_lesser_or_equal: 0
  address_required: 0
  email_required: 0
  in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
  max_individual_eligibility: 2
  itemArray [
    {
      purchase_item_id: 23329
      purchase_item_code: 850016300034
      purchase_item_description: Primal – Pork Butchers Blend Topper 2lb
      purchase_item_rebate_value: 2.00
    }
    {
      purchase_item_id: 23325
      purchase_item_code: 850016300171
      purchase_item_description: Primal – Chicken Market Mix Topper 5lb
      purchase_item_rebate_value: 5.00
    }
  ]
}
```

BuyAnyXGetAnyOtherY – with this type, there is a group of purchase items and a group of redeemable items. You have to purchase the required purchase amount from the purchase group to get the free qty amount from the free group

```
{
  astro_mfg_id: 148
  astro_mfg_name: Grizzly Pet Products
  astro_mfg_program_id: 7713
  astro_mfg_program_title: Grizzly Pet | Buy any Joint Aid, get 11lb bag free
  astro_program_type: offer
  program_start_date: 2020-07-01
  program_end_date: 2020-07-31
  program_calculation_type: BuyAnyXGetAnyOtherY
  redemption_lesser_or_equal_value: 0
  address_required: 0
  email_required: 1 (this means email will be required for redemption)
  in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
  required_purchase_amount: 1
  free_qty: 1
  max_individual_eligibility: 2
  itemArray: [
    {
      purchase_item_id: 10579
      purchase_item_code: 835953005426
      purchase_item_description: Grizzly JOINT AID – Pellet
      purchase_item_value: 40
    }
    {
      purchase_item_id: 20863
      purchase_item_code: 835953008687
      purchase_item_description: Grizzly JOINT AID – Hemp Enhanced Pellet
      purchase_item_value: 53
    }
  ]
  freeItemArray: [
    {
      free_item_id: 10565
      free_item_code: 835953009059
      free_item_description: Grizzly Superfoods – Salmon – Dog Food
      free_item_value: 11
    }
    {
      free_item_id: 10567
      free_item_code: 835953009103
      free_item_description: Grizzly Superfoods – Salmon – Cat Food
      free_item_value: 11
    }
  ]
}
```

```
}
```

BuyXGetYAnyProduct – with this type the purchase product pool is the same as the free product pool. The customer must purchase the required number of items and then they get the free qty of items from the same list for free.

```
{
```

```
    astro_mfg_id: 237
```

```
    astro_mfg_name: Boss Dog Brands
```

```
    astro_mfg_program_id: 7749
```

```
    astro_mfg_program_title: Boss Dog Brands | Buy 2 Yogurt Singles, Get 1 Free
```

```
    astro_program_type: offer
```

```
    program_start_date: 2020-07-01
```

```
    program_end_date: 2020-07-31
```

```
    program_calculation_type: BuyXGetYAnyProduct
```

```
    redemption_lessor_or_equal_value: 1
```

```
    address_required: 0
```

```
    email_required: 0
```

```
    in_store_only: 0 (if value is 1, program is not to be used in eCommerce)
```

```
    required_purchase_amount: 2
```

```
    free_qty: 1
```

```
    max_individual_eligibility: 1
```

```
    itemArray: [
```

```
        {
```

```
            purchase_item_id: 17087
```

```
            purchase_item_code: 860379002346
```

```
            purchase_item_description: Boss Dog – Greek Style Frozen Yogurt – Pumpkin
```

```
            purchase_item_value: 2
```

```
        }
```

```
        {
```

```
            purchase_item_id: 17086
```

```
            purchase_item_code: 860379002315
```

```
            purchase_item_description: Boss Dog – Greek Style Frozen Yogurt – Cheddar and Bacon
```

```
            purchase_item_value: 2
```

```
        }
```

```
    ]
```

```
}
```

addOfferRedemptionOnly

The addOfferRedemptionOnly API allows offers to be redeemed after the sale is complete. It only requires the customerID, program id and the UPCs that were redeemed. Each customer id / program id combination has to make an API call, but multiple redemptions for the same customer / program combination. For each redemption, create a new sub array in the redemption array. Some programs allow for 2 or more free items per redemption. In those cases, the item sub array will have 2 or more elements. For rebates, the item passed needs to be the item the customer received the rebate on. If possible, please include the total msrp (or purchase price) for each redemption of the items purchased to complete the offer.

URL: <https://api.astroloyalty.com/api/json/addOfferRedemptionOnly/>

Parameters:

- jsonData={
 "customerID": [internal customer id]
 "astro_program_id": [offer program id for the redemption]
 "redemptionArray": {
 itemArray [{
 item_code: [first redemption: upc being redeemed]
 }
]
 offer_purchase_msrp: 15.00
 itemArray [{
 item_code: [second redemption: upc being redeemed]
 }
]
 offer_purchase_msrp: 27.00
}

Return Example

```
(  
  astro_status: 100  
  astro_status_message: Success  
  returnData [  
    {  
      redemption:0  
      redemption_status_code: 100  
      redemption_status_message: Success  
      astro_reward_id: 1231881  
    }  
    {  
      redemption:1  
      redemption_status_code: 755  
      redemption_status_message: Max Eligibility Reached  
    }  
  ]  
)
```

Loyalty Points Endpoints

customerPointsStatus

The customerStatus API returns the customers current loyalty points status, loyalty points transactions and whether or not they are eligible for points redemption.

URL: <https://api.astroloyalty.com/api/json/customerPointsStatus/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
}

Return Example

```
{
  status:100
  status_message:Success
  returnData: {
    customerID:11
    first_name:Mark
    last_name:Keeran
    astroPointsBalance: 525
    astroPointsTransactions:[
      {
        astro_points_transaction_id: 131881
        pointsDescription: Earn 1 point per dollar
        pointsDate: 2018-11-05
        pointsQty: 25
        totalPoints: 25
      }
      {
        astro_points_transaction_id: 135871
        pointsDescription: Earn 1 point per dollar
        pointsDate: 2018-11-24
        pointsQty: 500
        totalPoints: 500
      }
    ]
    astroPointsEligibleRewards:[
      {
        astro_points_reward_id: 1214
        pointsRewardTitle: Earn 300 points, Get $15 off
        pointsRequired: 300
        reward_type: rebate
        reward_value: 15
      }
    ]
  }
}
```

addPointsByDollar

The addPointsByDollar API allows you to add customer loyalty points to the Astro platform based on the purchase dollars of the transaction. Astro will determine the correct points value based on the Loyalty Event setup by the retailer in Astro.

URL: <https://api.astroloyalty.com/api/json/addPointsByDollar/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "dollarAmount": [dollar amount associated with the sale]
 "astro_points_event_id": [**Optional** – If the retailer has multiple points events, the points can be assigned directly. If let out, it will use the default RestAPI event]
}

Return Example

```
{
  status: 100
  status_message: Success
  returnData: {
    astro_points_transaction_id: 131381
    astro_points_event_id: 1351
    total_points: 25
  }
}
```

addPoints

The addPoints API allows you to add customer loyalty points to the Astro platform in any point amount you want.

URL: <https://api.astroloyalty.com/api/json/addPoints/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "totalPoints":[amount of points to add]
 "astro_points_event_id":[**Optional** – If the retailer has multiple points events, the points can be assigned directly. If let out, it will use the default RestAPI event]
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: {  
    astro_points_transaction_id: 131381  
    astro_points_event_id: 1351  
    total_points: 25  
  }  
}
```

redeemPoints

The redeemPoints API allows you to add redeem an eligible point reward for a customer. You only need to specify the customer and the astro_points_reward_id (which is provided via the getCustomStatus or getCustomerPointsStatus). This **only** deducts points from the Astro program. You still need to apply the appropriate discount or cash back amount to the sale.

URL: <https://api.astroloyalty.com/api/json/redeemPoints/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "astro_points_reward_id":[ID provided from the getCustomerStatus or getCustomerPointsStatus for eligible points redemptions]
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: {  
    astro_points_transaction_id: 131381  
    astro_points_reward_id: 312  
    total_points: -300  
  }  
}
```


addPointsByPurchaseBatch

The addPointsByPurchaseBatch API allows you to add customer loyalty points to the Astro platform based on the purchase dollars of each sku on the sale. The benefit of this option is allowing the retailer to create special points scenarios based on products purchased. Astro will determine the correct points value based on the Loyalty Event setup by the retailer in Astro.

URL: <https://api.astroloyalty.com/api/json/addPointsByPurchaseBatch/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "transactions": [
 {
 "transactionID": [internal unique transaction id]
 "item_code": [UPC of the purchased item]
 "item_qty": [optional, if not present 1 is assumed]
 "item_subtotal": [subtotal of the item minus discounts and tax]
 }
 {
 "transactionID": [internal unique transaction id]
 "item_code": [UPC of the purchased item]
 "item_qty": [optional, if not present 1 is assumed]
 "item_subtotal": [subtotal of the item minus discounts and tax]
 }
]
}

Return Example

```
{  
  status: 100  
  status_message: Success  
  returnData: {  
    astro_points_transaction_id: 131381  
    astro_points_event_id: 1351  
    total_points: 25  
  }  
}
```

Coupon Endpoints

Astro provides a coupon service for our MFG partners that allows them to target certain pet owner with specific deals. While the deals can be similar to an Astro Offer, unlike offers, they are not available to all pet owners. If a pet owner has been issued a coupon, they will receive an email notification and the customerStatus endpoint will include an additional section for coupons. This will tell you that they are eligible for a coupon redemption and the relevant information about the coupon. The required_purchase_before_redemption option indicates whether the item must be added via the addTransaction endpoint prior to redeeming the coupon. The data will appear as follows:

```
customerCoupons: [
  {
    coupon_code: AC0000228
    coupon_title: Best Dog and Cat | Dollars Off Food
    reward_type: rebate
    coupon_expire_date: 2020-05-31
    required_purchase_before_redemption: true
    coupon_items: [
      {
        astro_item_id: 45
        item_code: 12341234123
        item_description: Happy Cat Natural Beef – 10lb
        rebate_amount: 10.00
      }
      {
        astro_item_id: 46
        item_code: 12341234125
        item_description: Happy Cat Natural Beef – 5lb
        rebate_amount: 5.00
      }
    ]
  }
]
```

If the customer is eligible for a coupon, you can use the below endpoints to complete the coupon redemption in Astro.

addCouponRedemption

The addCouponRedemption API allows you to redeem an eligible coupon for a customer. Remember that if required_purchase_before_redemption is true, the item must be first added to the sale via the addTransaction endpoint before adding the coupon redemption.

URL: <https://api.astroloyalty.com/api/json/addCouponRedemption/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "coupon_code": [coupon code provided in getCustomerStatus
 "astro_item_id": [ID provided from the getCustomerStatus for eligible coupon items]
}

Return Example

```
{  
  status:100  
  status_message:Success  
  returnData: {  
    rebate_amount: 10.00  
  }  
}
```

removeCouponRedemption

The removeCouponRedemption API allows you to reset a coupon redemption if the customer chooses not to apply the coupon or decides to apply it to a different purchased item. A coupon can only be applied once, so this allows you to reset the coupon application.

URL: <https://api.astroloyalty.com/api/json/removeCouponRedemption/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "coupon_code": [coupon code provided in getCustomerStatus
}

Return Example

```
{  
  status:100  
  status_message:Success  
}
```

Marketing Endpoints

sendEmail

The sendEmail API allows you to send the customer the Welcome/Verification email. It provides links to the Astro mobile app as well as an option to set their mobile app password.

URL: <https://api.astroloyalty.com/api/json/sendEmail/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
}

Return Example

```
{  
    status:100  
    status_message:Success  
}
```

sendSMS

The sendSMS API allows you to send the customer the opt in SMS. This feature is only available to retailers who subscribe to the Astro Marketing Suite and SMS text messaging services. If the customer has already opted in for SMS or has opted out, the SMS opt in will not be sent.

URL: <https://api.astroloyalty.com/api/json/sendSMS/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
}

Return Example

```
{  
    status:100  
    status_message:Success  
}
```

listCustomerTags

The listCustomerTags API endpoint provides a list of all the of the custom Customer Tags the retailer has established in Astro

URL: <https://api.astroloyalty.com/api/json/listCustomerTags/>

Parameters:

None

Return Example

```
{
  status:100
  status_message:Success
  returnData: {
    {
      tag_id: 145
      tag_name: VIP
    }
    {
      tag_id: 147
      tag_name: Military
    }
  }
}
```

addCustomerTag

The addCustomerTag API endpoint allows for associating an existing customer tag with an existing customer account.

URL: <https://api.astroloyalty.com/api/json/addCustomerTag/>

Parameters:

- jsonData = {
 "customerID":[internal customer id]
 "tag_id": [tag_id]
}

Return Example

```
{
  status:100
  status_message:Success
}
```

deleteCustomerTag

The deleteCustomerTag API endpoint allows for the removing of an associated customer tag from an existing customer account.

URL: <https://api.astroloyalty.com/api/json/removeCustomerTag/>

Parameters:

- jsonData = {
 "customerID": [internal customer id]
 "tag_id": [tag_id]
}

Return Example

```
{  
    status:100  
    status_message:Success  
}
```

Status Codes

Astro_Status	Astro_Status_Message	Description
100	Success	Successful API Call
200	Hourly Rate Limit Exceeded	More API calls than allowed in one hour, need to wait until next hour to continue
220	Invalid API Credentials	Username, Password or Client ID are invalid
250	Invalid Token	Token sent for Authorization is invalid
260	Invalid Client ID	Client ID is missing or is invalid
300	Token Expired	Token sent for Authorization has expired
310	No Customer Found	No matching customer for customerID
315	Customer Account has been combined with different customer	The customerID linked account has been rolled up to another customer account
320	Customer Account Already Linked	Internal Customer ID already linked to Astro Customer ID
330	Email or Phone Already in Use	Email address or phone number is in use by different customer
340	Email address is invalid	The email address provided is not valid
400	Incomplete Request	Not all required parameters were sent for API request
410	No Program Found	Program ID supplied does not exist or is not enrolled
510	TransactionID Already Exists	transactionID has already been passed to Astro
520	transactionID Does not Exist	transactionID does not exist in Astro
550	No Item Found	itemCode does not exist in Astro
560	Program No Longer Available	Program has been discontinued or sunset
575	Undetermined Loyalty Event	Either the event id that was passed was incorrect or no event id was passed and Astro could not determine the default event
600	Cannot Delete Card has been redeemed	transactionID is on card that was redeemed and therefore cannot be deleted
625	Customer Already Exists	customerID is already in Astro
650	Multiple Program Error	Item exists on more than one program. Contact Astro Support
660	No Eligible Coupon Purchase	Coupon being redeemed has no eligible purchase items for date of redemption.
665	Invalid Coupon Item	Coupon item being redeemed is invalid
670	Invalid Coupon Code	Coupon code is not a valid or active coupon
675	Coupon Code Redeemed	Coupon has already been redeemed
700	No Reward Found	astro_reward_id does not exist
710	Customer Address Required	Redemption Request missing required customer information
720	Customer Email Required	Redemption Request missing required customer information
730	Ineligible Reward Item	item_id passed for redemption request is ineligible per program rules

740	Reward Already Redeemed	Redemption transmitted for a reward that has already be redeemed
745	Offer Reward has Expired	Offer Reward has Expired
750	Reward Marked as Fulfilled	Reward has been fulfilled by distribution and cannot be removed
755	Max Eligibilty Reach	Offer or customer has reached the maximim eligibility
760	Ineligible Points Reward ID	Points Reward ID passed is not eligible for redemption
770	Offer Program ID required	Item is eligible for more than one offer so offer program choice is required
775	Incorrect Redemption Quantity	The number of offer UPCs sent do not match the free quantity of the offer.
780	Invalid Customer Email Address	Customer has invalid email address and can't be sent Welcome Email
790	SMS Not enabled	SMS is not active for retail account
795	SMS Not sent	The customer has already opted in or has opted out of SMS
900	Invalid internalLocationID	The internalLocationID is not valid or not mapped to Astro location
1000	Unknown error	An unkown error has occurred. Contact Astro Support